ASSIGNMENT - 4

① $$u' = \frac{u - v}{1 - \frac{uv}{c^2}} \quad \text{— ①}$$

$u, v < c \Rightarrow \boxed{\frac{u}{c}, \frac{v}{c} < 1}$

Let $\frac{u}{c} = x$; $\frac{v}{c} = y$; $\frac{u'}{c} = z$

$\Rightarrow \boxed{x, y < 1}$

① can be written as :

$$z = \frac{x - y}{1 - xy}$$

Let $x = \frac{1-\alpha}{1+\alpha}$ & $y = \frac{1-\beta}{1+\beta}$; $\alpha, \beta \in R^+$

$$\therefore z = \frac{\frac{1-\alpha}{1+\alpha} - \frac{1-\beta}{1+\beta}}{1 - \frac{(1-\alpha)(1-\beta)}{(1+\alpha)(1+\beta)}}$$

$$= \frac{(1-\alpha)(1+\beta) - (1-\beta)(1+\alpha)}{(1+\alpha)(1+\beta) - (1-\alpha)(1-\beta)}$$

$$= \frac{\cancel{1} + \beta - \alpha - \cancel{\alpha\beta} - \cancel{1} - \alpha + \beta + \cancel{\alpha\beta}}{\cancel{1} + \alpha + \beta + \cancel{\alpha\beta} - \cancel{1} + \beta + \alpha - \cancel{\alpha\beta}}$$

$$= \frac{2(\beta - \alpha)}{2(\alpha + \beta)} = \frac{\beta - \alpha}{\beta + \alpha} = 1 - \frac{2\alpha}{\alpha + \beta}$$

$$\therefore z = 1 - \frac{2\alpha}{\alpha + \beta} < 1$$

② $$\eta = \frac{1}{2} \ln\left(\frac{1+\beta}{1-\beta}\right)$$

$$\Rightarrow 2\eta = \ln\left(\frac{1+\beta}{1-\beta}\right)$$

$$\Rightarrow e^{2\eta} = \frac{1+\beta}{1-\beta} \Rightarrow \beta = \frac{e^{2\eta} - 1}{e^{2\eta} + 1} = \tanh(\eta)$$

$$\gamma = \frac{1}{\sqrt{1-\beta^2}} = \cosh(\eta)$$

$$\beta = \frac{v}{c} = \tanh(\eta) \; ; \quad \gamma\beta = \cosh(\eta) \cdot \tanh(\eta)$$
$$= \sinh(\eta)$$

$$\cancel{\therefore \; x' = \frac{x - vt}{\sqrt{1-\frac{v^2}{c^2}}} = [x - c\tanh(\eta)\, t]\cosh(\eta)}$$

$$\cancel{y' = y \; ; \quad z' = z}$$

$$\cancel{t' = \frac{t - \frac{vx}{c^2}}{\sqrt{1-\frac{v^2}{c^2}}} = \left[t - \frac{\tanh(\eta)\, x}{c}\right]\cosh(\eta)}$$

$$\begin{bmatrix} t' \\ x' \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix}$$

$$\cancel{\begin{bmatrix} x' \\ t' \end{bmatrix} = \begin{bmatrix} \cosh \end{bmatrix}}$$

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = \begin{bmatrix} \cosh(\eta) & -\sinh(\eta) \\ -\sinh(\eta) & \cosh(\eta) \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix}$$

For small η ;

$$\sinh(\eta) = \eta$$
$$\cosh(\eta) = 1$$

$$\begin{bmatrix} x' \\ t' \end{bmatrix} = \begin{bmatrix} 1 & -\eta \\ -\eta & 1 \end{bmatrix} \begin{bmatrix} x \\ t \end{bmatrix}$$

③ $$\begin{bmatrix} t' \\ x' \end{bmatrix} = \begin{bmatrix} \cosh(\eta_1) & -\sinh(\eta_1) \\ -\sinh(\eta_1) & \cosh(\eta_1) \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix}$$

$$\begin{bmatrix} t'' \\ x'' \end{bmatrix} = \begin{bmatrix} \cosh(\eta_2) & -\sinh(\eta_2) \\ -\sinh(\eta_2) & \cosh(\eta_2) \end{bmatrix} \begin{bmatrix} t' \\ x' \end{bmatrix}$$

$$= \begin{bmatrix} \cosh(\eta_2) & -\sinh(\eta_2) \\ -\sinh(\eta_2) & \cosh(\eta_2) \end{bmatrix} \begin{bmatrix} \cosh(\eta_1) & -\sinh(\eta_1) \\ -\sinh(\eta_1) & \cosh(\eta_1) \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix}$$

$$= \begin{bmatrix} \cosh(\eta_1+\eta_2) & -\sinh(\eta_1+\eta_2) \\ -\sinh(\eta_1+\eta_2) & \cosh(\eta_1+\eta_2) \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix}$$

$$\begin{bmatrix} t' \\ x' \end{bmatrix} = \begin{bmatrix} \cosh \eta_3 & -\sinh \eta_3 \\ -\sinh(\eta_3) & \cosh \eta_3 \end{bmatrix} \begin{bmatrix} t \\ x \end{bmatrix}$$

where $\eta_3 = \eta_1 + \eta_2$ = rapidity of Charlie wrt Bob.